

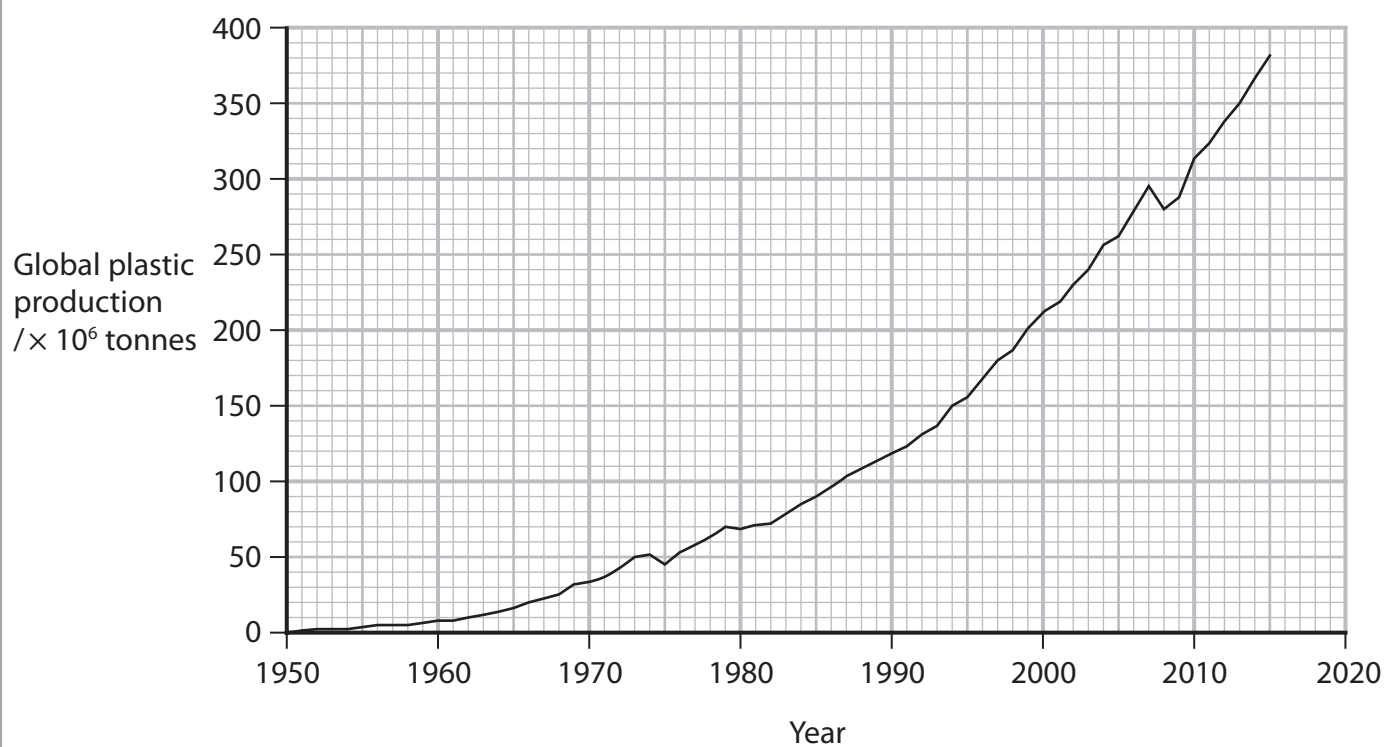
9 The photograph shows a sea turtle.



(Source: © M Swiet Productions/Getty Images)

Sea turtles can take in plastic debris when they eat. This plastic can harm the turtles.

(a) The graph shows the global production of plastics per year from 1950 to 2015.



(i) Calculate the mean rate of global plastic production from 2000 to 2015.

Give your answer to 3 significant figures.

(2)

Answer million tonnes per year

(ii) Estimate the global plastic production in 2023.

Use your calculated value from part (i).

(1)

Answer million tonnes

(iii) Suggest why this value may be an underestimate.

Use the information in the graph to support your answer.

(1)

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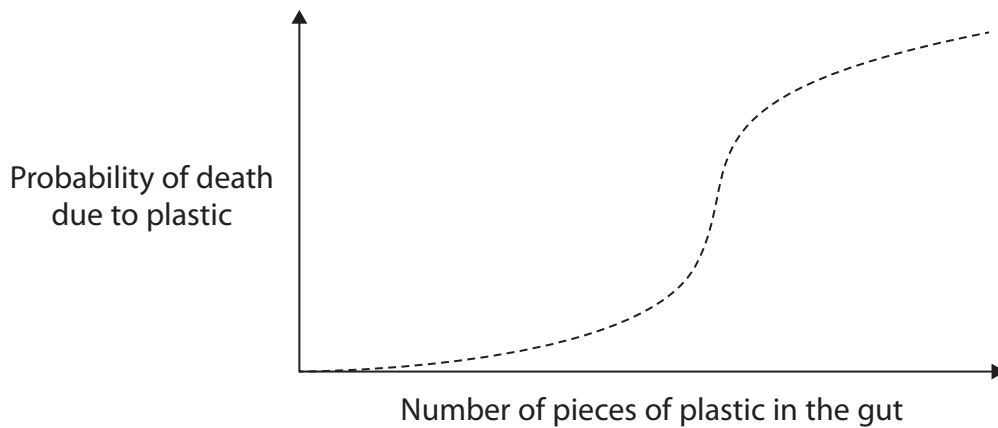
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(b) One group of scientists determined the number of pieces of plastic in the guts of individual sea turtles.

They then modelled this data to estimate the probability of death due to plastic.

The graph shows the results of this modelling.



(i) Describe how the probability of death due to plastic increases with the number of pieces of plastic in the gut.

(2)

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(ii) Describe **two** limitations of using this graph to determine the probability of death due to the ingestion of plastic.

(2)

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*(c) Mangoes are tropical fruits that are harvested before they are ripe.

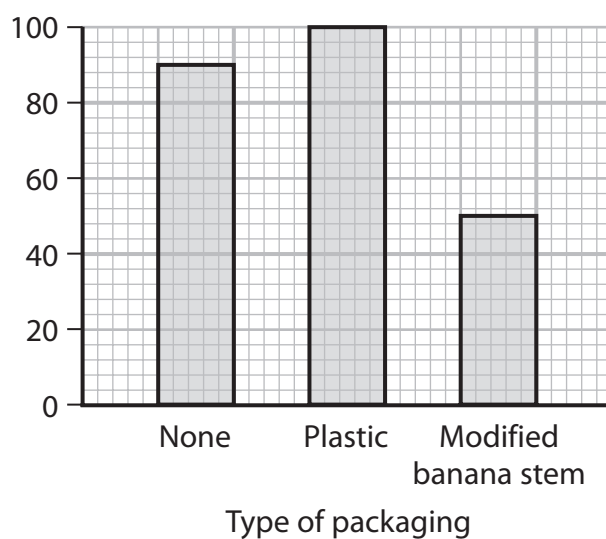
Plastic packaging is used to wrap mangoes after they have been picked. This preserves the fruit during transportation and storage.

Banana stalks have a high content of cellulose and have been modified to use as packaging material for mangoes.

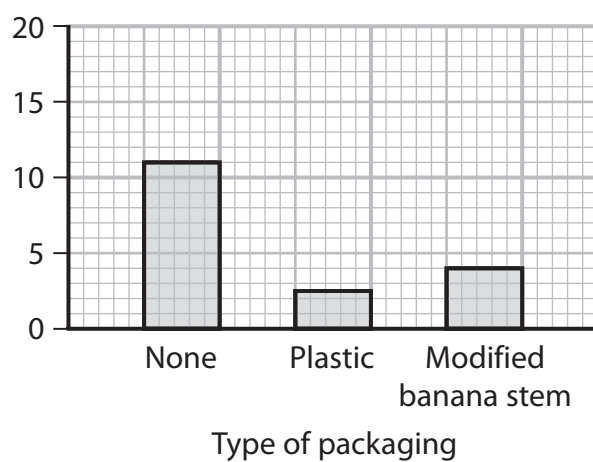
A study compared the use of these two types of material and their decomposition.

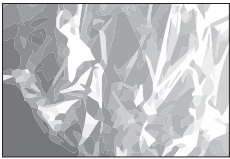
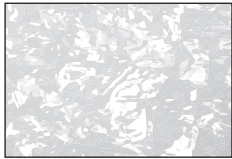
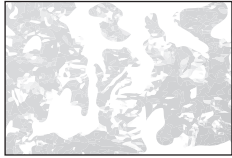
The graphs and the table show the results of this study.

Percentage of mangoes that are diseased (%)



Percentage of total mass of mangoes lost (%)



Time of decomposition of packaging / weeks	Appearance of packaging	
	Plastic packaging	Modified banana packaging
0		
2		
4		

Use the information in the graphs, the table and your own knowledge to support your answer.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

TOTAL FOR PAPER = 90 MARKS